

Why Sorting?

- Obvious applications
 - Organize an MP3 library
 - Maintain a telephone directory
- Problems that become easy once items are in sorted order
 - Find a median, or find closest pairs
 - Binary search, identify statistical outliers
- Non-obvious applications
 - Data compression: sorting finds duplicates
 - Computer graphics: rendering scenes front to back

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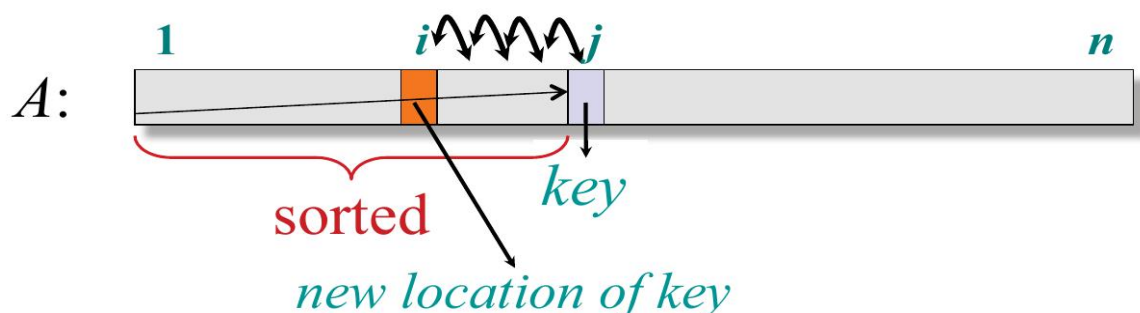
Insertion sort

INSERTION-SORT (A, n) $\triangleright A[1 \dots n]$

for $j \leftarrow 2$ to n

 insert key $A[j]$ into the (already sorted) sub-array $A[1 \dots j-1]$.
 by pairwise key-swaps down to its right position

Illustration of iteration j



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Binary Insertion sort

BINARY-INSERTION-SORT (A, n) $\triangleright A[1 \dots n]$
for $j \leftarrow 2$ to n
 insert key $A[j]$ into the (already sorted) sub-array $A[1 \dots j-1]$.
 Use binary search to find the right position

Binary search with take $\Theta(\log n)$ time.

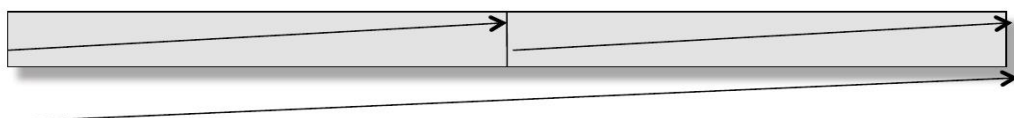
However, shifting the elements after insertion will still take $\Theta(n)$ time.

Complexity: $\Theta(n \log n)$ comparisons
 (n^2) swaps

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Meet Merge Sort

divide and conquer { **MERGE-SORT** $A[1 \dots n]$
 1. If $n = 1$, done (nothing to sort).
 2. Otherwise, recursively sort $A[1 \dots n/2]$ and $A[n/2+1 \dots n]$.
 3. “**Merge**” the two sorted sub-arrays.



Key subroutine: MERGE

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Analyzing merge sort

MERGE-SORT $A[1 \dots n]$

- | | |
|---|-------------|
| 1. If $n = 1$, done. | $T(n)$ |
| 2. Recursively sort $A[1 \dots \lceil n/2 \rceil]$ and $A[\lceil n/2 \rceil + 1 \dots n]$. | $\Theta(1)$ |
| 3. “Merge” the two sorted lists | $2T(n/2)$ |
| | $\Theta(n)$ |

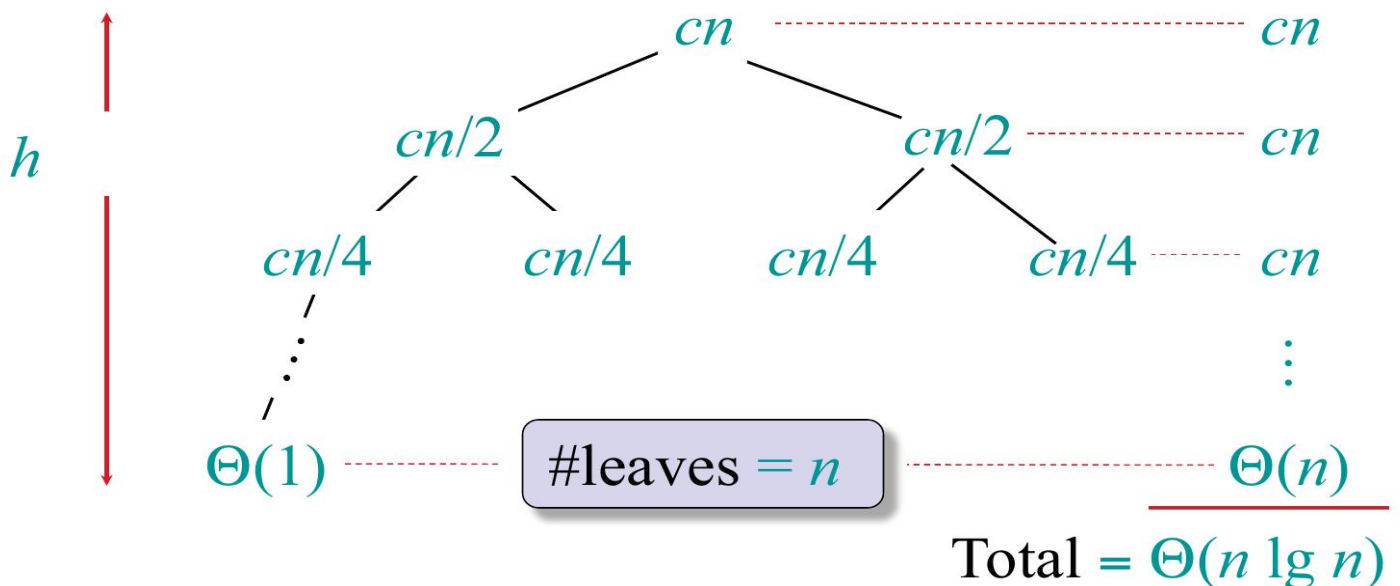
$$T(n) = \begin{cases} \Theta(1) & \text{if } n = 1; \\ 2T(n/2) + \Theta(n) & \text{if } n > 1. \end{cases}$$

$$T(n) = ?$$

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Recursion tree

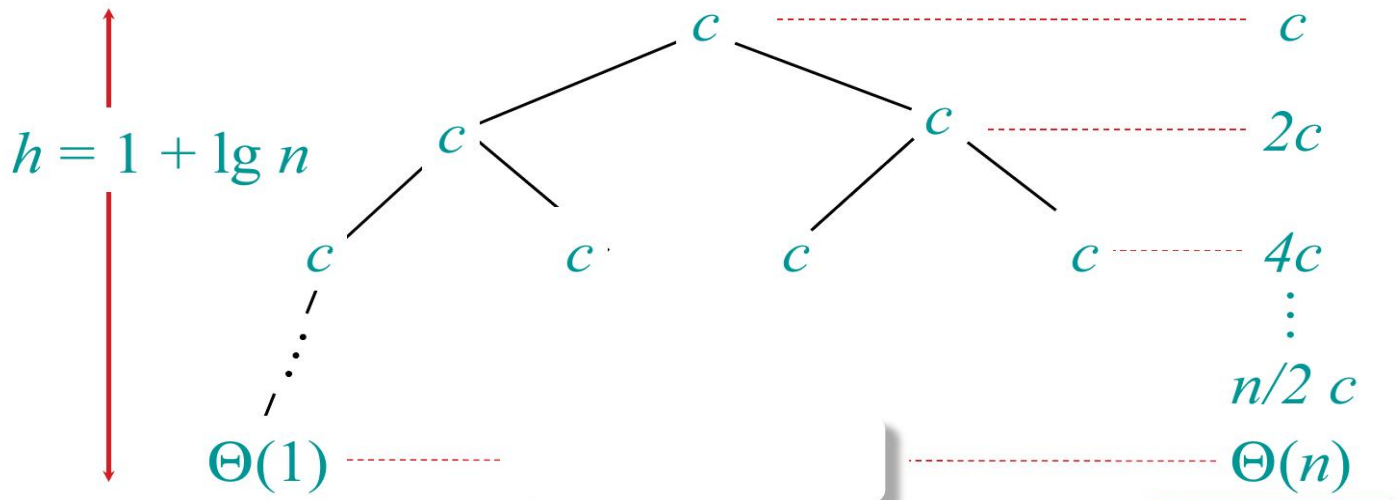
Solve $T(n) = 2T(n/2) + cn$, where $c > 0$ is constant.



Equal amount of work done at each level

Tree for different recurrence

Solve $T(n) = 2T(n/2) + c$, where $c > 0$ is constant.



Note that $1 + \frac{1}{2} + \frac{1}{4} + \dots < 2$

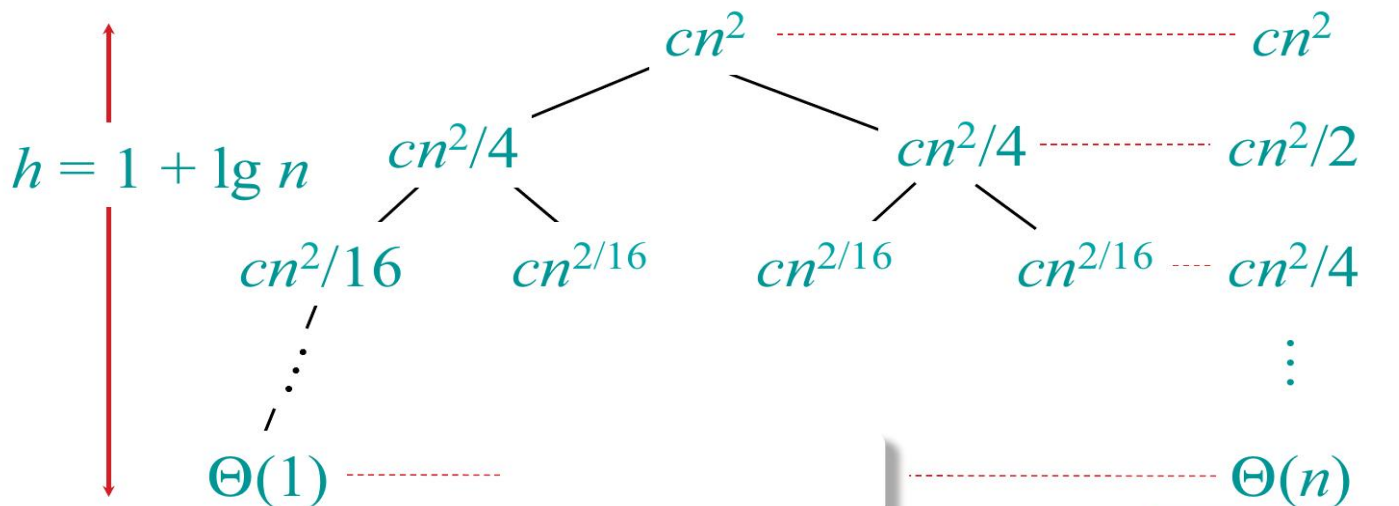
All the work done at the leaves

Total = $\Theta(n)$

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Tree for yet another recurrence

Solve $T(n) = 2T(n/2) + cn^2$, $c > 0$ is constant.



Note that $1 + \frac{1}{2} + \frac{1}{4} + \dots < 2$

All the work done at the root

Total = $\Theta(n^2)$

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